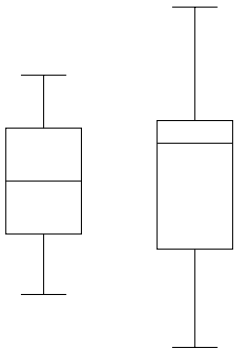
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DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date : 01/04/2025	**CIE – I**	**Max. Marks : 10 + 50**
Semester : IV	**UG**	**Duration : 30 + 90 Min**
Course Title: STATISTICS FOR DATA ANALYTICS		**Course Code : IM241AT**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1.	40th percentile in a given set of 140 data values occurs at position _____.	1	2	1
2.	<table><tr><td>Class</td><td>21”–30”</td><td>31”–40”</td><td>41”–50”</td><td>51”–60”</td><td>61”–70”</td></tr><tr><td>No of occurrences</td><td>12</td><td>16</td><td>22</td><td>25</td><td>18</td></tr></table> The median in the above data lies in the class _____.	1	2	1
3.	According to Bayes' rule, P(A|B) = _____, for two events in a sample space S.	1	1	1
4.	X = the number coming up when a 6-faced die is thrown. Y = number of heads when a coin is tossed. The possible values of X − Y are _____.	1	3	2
5.	A = {1, 2, 3, 4}; B = {2, 4, 6, 8}; C = {6, 7, 8, 9}; D = {4, 5, 6, 7}. If the sample space S = A ∪ B ∪ C ∪ D, and each element of S is equally likely to happen, then P(A′∩B) = _____.	1	3	2
6.	P(X ≤ 3 | Y = 0) is _____, for X and Y as in Question 4.	1	3	2
7.	A lot of 100 semiconductor chips contains 25 that are defective. Two are selected randomly, without replacement, from the lot. What is the probability that both are defective?	2	3	1
8.	Cooking oil is produced in two main varieties: mono and polyunsaturated. Two common sources of cooking oil are corn and canola. The following table shows the number of bottles of these oils at a supermarket: <table><tr><td rowspan="2">Type of unsaturation</td><td colspan="2">Type of oil</td></tr><tr><td>Canola</td><td>Corn</td></tr><tr><td>Mono</td><td>7</td><td>13</td></tr><tr><td>Poly</td><td>93</td><td>77</td></tr></table> If a bottle of oil is selected at random, what is the probability that it belongs to the polyunsaturated category?	2	3	1
	Part – B			

1.	Fourteen measurements of furnace temperatures (in °F) are recorded on successive batches in a semiconductor manufacturing process as follows: 953, 950, 948, 955, 951, 949, 957, 954, 955, 949, 952, 931, 953, 955. a) Calculate sample mean, sample variance, and standard deviation. b) Find the mode and median. How much could the largest temperature measurement be increased without changing the median value? c) Construct a box plot of the data, showing clearly the outlier bounds. d) Comment on the symmetry of the data.	10	3	1
2.	**a)** Describe any similarities or differences in the two distributions represented by the following boxplots. Assume the two boxplots have the same scale. [5 marks]  **b)** In a certain factory, machines 1, 2, and 3 are all producing pins. Of their production, machines 1, 2, and 3 produce 2%, 1%, and 3% defective pins respectively. Of the total production of pins, machine 1 produces 35%, machine 2 produces 25% and machine 3 produces 40%. If one pin is selected at random from the total pins, what is the probability that it is defective and find the conditional probability that it was produced by machine 3. [5 marks]	10	4	1
3.	Customers evaluate products and either give them good reviews or don't. Also, the products are highly successful, moderately successful or poor according to sales performance. In the past, 95% of highly successful products received good reviews, 60% of moderately successful products received good reviews, and 10% of poor products received good reviews. In addition, 40% of products have been highly successful, 35% have been moderately successful, and the rest have been poor products. a) Depict the sample space with appropriate partitions in a sketch. What is the probability that a product attains a good review? [4] b) If a product attains a good review, what is the probability that it will be a highly successful product? [3] c) If a product does not attain a good review, what is the probability that it will be a highly successful product? [3]	10	3	2
4.	Assume that a box contains 10 balls distributed as follows: Three are colored (C) & dotted (D); One is colored (C) & stripped (S); Two are gray (G) & dotted (D); Four are gray (G) & stripped (S). Find P(D|C), P(D|G), P(G|D), P(C|S), P(G).	10	3	2

5.	Suppose the excess length in a manufactured part in mm is recorded as below. 19 22 6 29 57 14 12 24 6 37 19 19 18 12 11 17 15 17 13 56 (i) Compute the mean, median and mode for the data set. (ii) Draw a histogram with 4 classes and a box plot, and interpret.	10	2	1

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Date : 18.06.2024	**CIE – I**	**Max. Marks : 50**
Semester : IV	**UG**	**Duration : 90 Min**
Course Title: STATISTICS FOR DATA ANALYTICS		**Course Code : IM241AT**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
1	Calculate Mean, Mode and Median for following group data Class Interval: Frequency 1 under 3: 4 3 under 5: 12 5 under 7: 13 7 under 9: 19 9 under 11: 7 11 under 13: 5	10	3	1
2	Construct a box plot for following data: 87 85 84 84 82 82 82 81 81 81 80 79 79 77 76 75 74 74 74 73 73 73 73 72 72 71 71 71 70 69 69 68 68 65 65 64 64 63 62 62	10	3	1
3	Assume the following are the marks obtained in Statistics for Data Analytics CIE 1. Calculate Mean, Median, Mode, Variance and Standard Deviation for the data. 33 43 44 45 46 49 31 32 34 36 39 26 29 22 22 23 25 11 19 18 19 15 23 45 44 29 39 38 28 27 26 28 35 39 38 38 37 37 28 29	10	3	1
4	Construct a Histogram for the following data. 35, 339, 650, 864, 1025, 1261, 38, 340, 655, 883, 1028, 1280, 48, 395, 669, 883, 1036, 1290, 53, 457, 703, 890, 1044, 1312, 70, 478, 730, 934, 1087, 1341, 99, 501, 763, 951, 1091, 1355, 138, 521, 788, 969, 1126, 1357, 164, 556, 789, 985, 1176, 1360, 220, 583, 789, 993, 1199, 1414, 265, 595, 802, 997, 1199, 1436, 272, 596, 822, 999, 1237, 1479, 312, 604, 851, 1018, 1242, 1492	10	3	2

5	Write a Note on Following (i) Population and Sample (ii) Parameter and Statistics (iii) Descriptive and Inferential Statistics (iv) Probability and Probability distribution	10	2	2

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Date : 5th July 2023	**CIE – I**	**Max. Marks : 50**
Semester : IV	**UG**	**Duration : 90 Min**
Course Title: STATISTICS FOR DATA ANALYTICS		**Course Code : 21IM41**

Note:
1. Answer all the Questions.

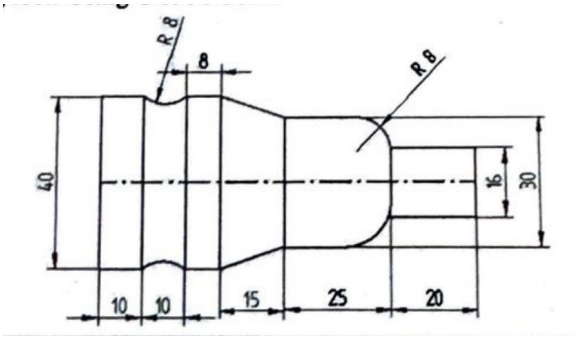
Sl. No	Questions	M	BT	CO
1.	20 weight measurements in a process were recorded as: 61 24 23 27 29 59 47 38 49 12 36 75 64 83 92 18 29 18 20 19 (i) Draw a box plot for the data. (ii) What are the range and the Inter Quartile Range for the data? (iii) Are there any outliers? If yes, what are they? (iv) Draw the stem and leaf diagram and comment on the distribution of data.	10	3	1
2.	Three machines A, B and C produce respectively 50%, 30% and 20% of total number of items in a factory. The proportion of defective output of these machines are 0.03, 0.04 and 0.05 respectively. If an item is selected at random find (i) The probability that the item selected is defective. (ii) If through an action of improvement, the percentage of defective output is reduced to 2%, 2% and 3% respectively. Revise the probability estimate of (i) above.	10	3	1
3.	A multiple-choice test contains 24 questions, each with four answers. Assume a student just guesses on each question. a) What is the probability that the student answers at least 21 questions correctly? [5] b) What is the probability the student answers at most 10 questions correctly? [5]	10	3	2
4.	Assume that a box contains 10 balls distributed as follows: Three are colored (C) & dotted (D); One is colored (C) & stripped (S); Two are gray (G) & dotted (D); Four are gray (G) & stripped (S). a) Find P(D|C), P(D|G), P(G|D), P(C|S), P(G)	10	3	2
5.	A random variable X is distributed as $f(x) = x/k$, $1 \leq x \leq 4$ a) What is the probability that X will take on a value of at most 2? [3] b) Plot the probability density function of X. [2] c) Derive the cumulative distribution of X and plot the same. [3] d) What is the median of X? Is it the same as mean of X? [2]	10	3	2

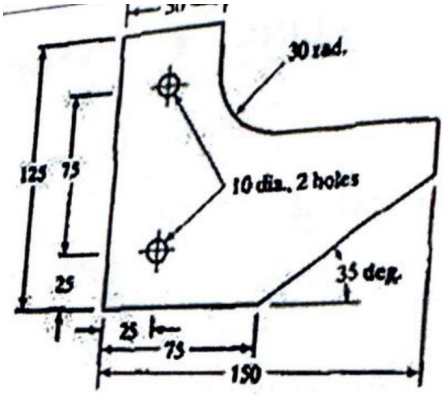
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Date : 1st April 2025	**CIE – I**	**Max. Marks : 10 + 50**
Semester : IV	**UG**	**Duration : 120 Min**
Course Title: CAD/CAM & ROBOTICS		**Course Code : IM343AI**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1.1	List any two advantages of using CAD over manual drafting?	2	2	1
1.2	How does DNC differ from CNC? State any two.	2	1	2
1.3	List any two reasons for implementing CAD in engineering and product design?	2	2	2
1.4	What are the key components of a DNC system?	2	2	1
1.5	What are two essential elements of an NC system?	2	1	2
	Part – B			
1	Discuss the various applications of computers in the design process with its various stages and their importance in engineering design.	10	1	2
2	According to the type of machine, NC machines are classified into three types. Explain the classification with the help of case examples.	10	2	1
3	Write a CNC turning cycle program for the given profile. The tool used is left hand cutting tool. Using G & M Codes. 	10	4	2

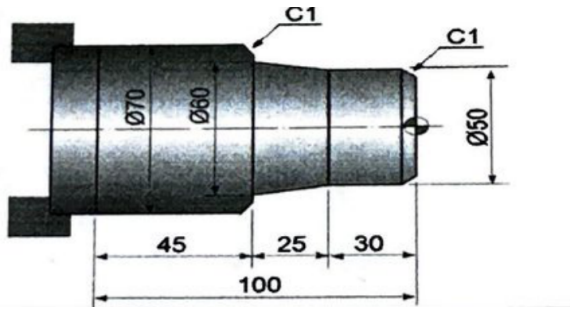
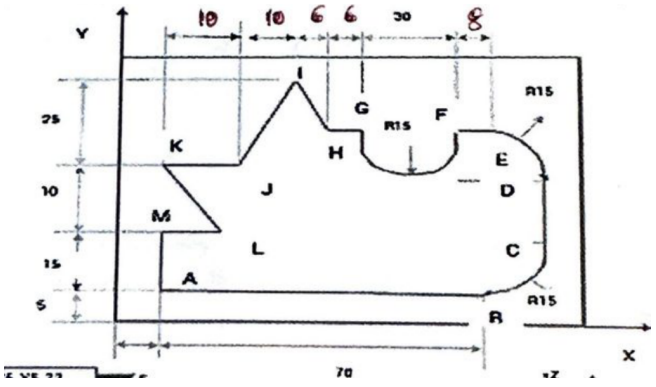
4	Develop a complete APT Part Program for machining the given part. (Profile Milling only). Tool – Slot drill Cutter of ϕ10 mm, Cutting Speed 1500, Feed 60, Depth of cut = 6 mm. 	20	4	1

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Date : 18th June 2024	**CIE – I**	**Max. Marks : 50**
Semester : IV	**UG**	**Duration : 90 Min**
Course Title: CAD/CAM & ROBOTICS		**Course Code : IM343AI**

Note:
1. Answer all the Questions.

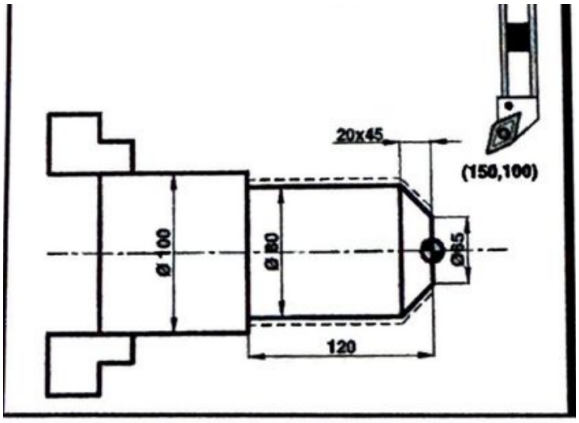
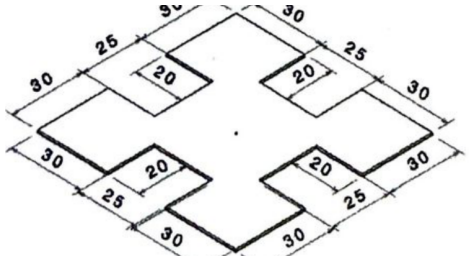
Sl. No	Questions	M	BT	CO
1	When is it appropriate to use the NC Program in manufacturing environment? Elaborate.	10	1	1
2	State the differences between the following? a) absolute and incremental coordinate system b) machine zero and program zero	10	2	1
3	Write a CNC turning cycle program for the given profile. The tool used is left hand cutting tool. C1 = 3 × 45°. 	15	3	1
4	Write a CNC Milling program for the given profile. The cutting tool used is slot drill cutter and the depth of cut is 5 mm. 	15	3	1

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Date : 6th July 2023	**CIE – I**	**Max. Marks : 50**
Semester : IV	**UG**	**Duration : 90 Min**
Course Title: CAD/CAM & ROBOTICS		**Course Code : 21IM43**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
1.	State the differences between Lathe, Numeric control machine and Computer numeric control with respect to quality, time, cost and automation?	10	1	1
2.	When is it appropriate to use the NC Program in manufacturing environment? Elaborate.	10	2	1
3.	Write a CNC turning cycle program for the given profile. The tool used is left hand cutting tool. 	15	4	2
4.	Write a CNC milling program for the given profile. The cutting tool used is slot drill cutter and the depth of cut is 5 mm. 	15	4	2

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Date : 02nd April 2025	**CIE – I**	**Max. Marks : 10 + 50**
Semester : IV	**UG**	**Duration : 30 + 90 Min**
Course Title: OPERATIONS RESEARCH		**Course Code : IM244AI**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1	The variables that represent decisions to be made in an optimization model are called _____.	1	1	1
2	In an optimization model, the function that needs to be maximized or minimized is called the _____.	1	1	1
3	Linear Programming is a mathematical technique used for _____ decision-making.	1	2	2
4	One of the key limitations of Operations Research is that it requires _____ data for accuracy.	1	2	2
5	The feasible region is the set of all possible solutions that satisfy all _____.	1	1	1
6	The graphical solution method of LPP is applicable only when there are _____ decision variables.	1	2	2
7	The additivity assumption ensures that the total effect of different decision variables is the _____ of their individual effects.	1	2	1
8	When the total supply does not equal total demand, the transportation problem is called a(n) _____ transportation problem.	1	2	2
9	A prohibited route transportation problem occurs when certain routes have _____ restrictions and cannot be used for transportation.	1	2	3
10	The General Mathematical Formulation of a transportation problem consists of an objective function, supply constraints, and _____ constraints.	1	2	2
	Part – B			
1	Describe the Seven-Step Model-Building Process in Operations Research with an example.	10	2	1
2	Explain the key features of Operations Research (OR) models and highlight its limitations.	10	2	1

<table>
<tr><td>3</td><td>Consider the following problem faced by a production planner of a soft drink plant. He has two bottling machines A and B. A is designed for 8-ounce bottles and B for 16-ounce bottles. However, each can also be used for both types of bottles with some loss of efficiency. The manufacturing data is as follows:
<table><tr><td>Machine</td><td>8-ounce Bottles</td><td>16-ounce Bottles</td></tr><tr><td>hline A</td><td>100/minute</td><td>40/minute</td></tr><tr><td>hline B</td><td>60/minute</td><td>75/minute</td></tr><tr><td>hline</td><td></td><td></td></tr></table>The machines can be run for 8 hours per day, 5 days per week. The profit on an 8-ounce bottle is Rs 1.5 and on a 16-ounce bottle is Rs 2.5. Weekly production of the drink cannot exceed 3,00,000 bottles and the market can absorb 25,000 8-ounce bottles and 7,000 16-ounce bottles per week. The planner wishes to maximize his profit, subject of course, to all the production and marketing restrictions. Formulate this problem as an LP model to maximize total profit.</td><td>10</td><td>3</td><td>2</td></tr>
<tr><td>4</td><td>A diet for a sick person must contain at least 4,000 units of vitamins, 50 units of minerals and 1,400 calories. Two foods A and B are available at a cost of Rs. 4 and Rs. 3 per unit, respectively. If one unit of A contains 200 units of vitamins, 1 unit of mineral and 40 calories and one unit of food B contains 100 units of vitamins, 2 units of minerals and 40 calories, formulate this problem as an LP model and solve it by graphical method to find the combination of foods to be used to have least cost.</td><td>10</td><td>3</td><td>2</td></tr>
<tr><td>5</td><td>Determine an initial basic feasible solution to the following transportation problem by using (a) NWCR, (b) LCM and (c) VAM.
<table><tr><td></td><td>D1</td><td>D2</td><td>D3</td><td>D4</td><td>Supply</td></tr><tr><td>hline S1</td><td>19</td><td>30</td><td>50</td><td>10</td><td>7</td></tr><tr><td>hline S2</td><td>70</td><td>30</td><td>40</td><td>60</td><td>9</td></tr><tr><td>hline S3</td><td>40</td><td>8</td><td>70</td><td>20</td><td>18</td></tr><tr><td>hline Demand</td><td>5</td><td>8</td><td>7</td><td>14</td><td></td></tr><tr><td>hline</td><td></td><td></td><td></td><td></td><td></td></tr></table></td><td>10</td><td>3</td><td>3</td></tr>
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Date : 19th June 2024	**CIE – I**	**Max. Marks : 50**
Semester : IV	**UG**	**Duration : 90 Min**
Course Title: OPERATIONS RESEARCH		**Course Code : IM244AI**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
1	In relation to the LP problem, explain the implications of the following assumptions of the model: (i) Linearity of the objective function and constraints, (ii) Continuous variables, (iii) Certainty.	10	1	1
2	The company manufactures the models A, B and C whose profit contributions per unit are Rs 20, Rs 50 and Rs 70 respectively. The minimum weekly production requirement is 30 units for Model A, 150 units for Model B and 60 units for Model C. Each type of tape recorder requires a certain amount of time to manufacture the component parts for assembly and packaging. Specifically, dozens of Model A units require 6 hours to manufacture, 5 hours to assemble, and 2 hours to package. The numbers corresponding to a dozen units of Model B are 5, 6 and 4 and for dozens of Model C units are 7, 8 and 5. Over the next week, the company provides 150 hours of manufacturing, 140 hours of assembly and 60 hours of packing time and packaging. Formulate this problem as an LP model for maximizing the company's gross profit.	10	2	2
3	Solve the following Linear Programming Problem by Graphical Method: Maximize Z = $300x_1 + 400x_2$ Subject to the constraints $5x_1 + 4x_2 \leq 200$ $3x_1 + 5x_2 \leq 150$ $5x_1 + 4x_2 \geq 100$ $8x_1 + 4x_2 \geq 80$ where $x_1, x_2 \geq 0$	10	3	3
4	There are four machines and four operators with their respective payoffs (in 100 rupees) as shown below. How the machines should be allocated to each operator so as to minimize the total pay-off. O_1 O_2 O_3 O_4 M_1 5 4 2 7 M_2 7 8 3 5 M_3 5 3 4 6 M_4 4 6 6 5	10	2	2

5	Five salesmen are to be assigned to five districts. Estimates of sales revenue (in thousands) for each salesman are given as follows: Find the assignment pattern that maximizes the sales revenue.	10	3	3

	A	B	C	D	E
1	32	38	40	28	40
2	40	24	28	21	36
3	41	27	33	30	37
4	22	38	41	36	36
5	29	33	40	35	39

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Date : 16th April 2026	**CIE – I (Test – 1)**	**Max. Marks : 50**
Semester : IV	**UG**	**Duration : 90 Min**
Course Title: MARKETING MANAGEMENT		**Course Code : IM345AT**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1	Differentiate between marketplace and market space.	2	L2	1
2	What is the first step of content marketing.	2	L2	1
3	Mention any two core concepts of marketing.	2	L1	2
4	Mention any four characteristics of a good content.	2	L1	1
5	Differentiate between push and pull marketing.	2	L2	2
	Part – B			
1	A newly launched health & wellness startup offering organic lifestyle products aims to build strong brand awareness through content marketing. Design a step-by-step content marketing strategy, clearly outlining: content formats (blogs, videos, podcasts, etc.), content themes, distribution platforms (owned, earned, paid media) and metrics for performance evaluation.	10	L3	1
2	A travel agency specializing in adventure tourism wants to improve its targeted marketing. How should they create a buyer persona, and what factors (e.g., demographics, interests, online behaviour) should they consider?	10	L4	3
3	A retail company has been using traditional marketing methods such as newspaper ads, billboards, and TV commercials. However, they are now considering a shift to digital marketing. Compare the benefits and limitations of both approaches and suggest a strategy for transitioning to digital marketing.	10	L3	1
4	How can businesses align Web Analytics KPIs with their organizational goals? Illustrate with examples.	10	L2	2
5	Explain the key components of Search Engine Optimization (SEO), including on-page, off-page, and technical SEO. How do these elements contribute to improving website ranking on search engines?	10	L2	2

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Date : 3rd April 2025	**CIE – I**	**Max. Marks : 10 + 50**
Semester : IV	**UG**	**Duration : 120 Min**
Course Title: MARKETING MANAGEMENT		**Course Code : IM345AT**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
	Part – A			
1	Differentiate between marketplace and market space.	2	2	1
2	What is the mantra of marketing?	2	2	1
3	Define affiliate marketing and give an example.	2	1	2
4	Mention any four characteristics of a good content.	2	1	1
5	Differentiate between push and pull marketing.	2	2	2
	Part – B			
1	A newly launched e-commerce company wants to build brand awareness through content marketing. Suggest a step-by-step strategy for them, including content types and distribution channels.	10	2	3
2	A travel agency specializing in adventure tourism wants to improve its targeted marketing. How should they create a buyer persona, and what factors (e.g., demographics, interests, online behaviour) should they consider?	10	4	3
3	A retail company has been using traditional marketing methods such as newspaper ads, billboards, and TV commercials. However, they are now considering a shift to digital marketing. Compare the benefits and limitations of both approaches and suggest a strategy for transitioning to digital marketing.	10	4	1
4	Identify and explain the various entities that can be marketed, providing relevant examples for each category.	10	3	2
5	How have the 4Ps of marketing (Product, Price, Place, and Promotion) evolved in the digital era? Provide examples of companies effectively leveraging digital strategies for each element.	10	3	4

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Date : 1.07.2024	**CIE – I**	**Max. Marks : 50**
Semester : IV	**UG**	**Duration : 90 Min**
Course Title: MARKETING MANAGEMENT		**Course Code : IM345AT**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
1	Describe the key principles of digital marketing and explain how they differ from traditional marketing approaches. Provide examples to illustrate your points.	10	2	1
2	Your company has a limited marketing budget and needs to choose between Search Engine Optimization, Pay Per Click, and Social Media Marketing to promote a new product. Based on the product being a high-tech gadget aimed at young professionals, which channel(s) would you prioritize and why? Justify your decision with specific examples.	10	3	1
3	Identify and explain at least five key metrics used to measure the impact and performance of content marketing efforts. How can these metrics influence future content strategies?	10	2	1
4	Imagine you are a digital marketing strategist at GreenTech Innovations, a company launching a new line of eco-friendly kitchen appliances called "EcoChef." Design a comprehensive content marketing strategy to promote these products. Include specific goals, types of content, distribution channels, and methods of engagement with your target audience.	10	3	1
5	Explain the importance of data-driven insights in social media marketing. Describe the process of developing audience insights using social media analytics tools. How can these insights inform the development and optimization of social media campaigns?	10	2	2

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Date : 6th July 2023	**CIE – I**	**Max. Marks : 50**
Semester : IV	**UG**	**Duration : 90 Min**
Course Title: MARKETING MANAGEMENT		**Course Code : 21IM44**

Note:
1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
1.	Explain the concept of market segmentation. Describe the bases for market segmentation.	10	2	1
2.	How digital marketing differs from traditional way of marketing. Highlight five key differences.	10	2	2
3.	How would you measure the performance of content marketing? Can you describe one example of good content you have experienced?	10	3	2
4.	Create a good content for any product of your choice. Take the help of infographics, information and quotes. Content should not exceed one page of your answer booklet.	10	3	3
5.	Analyse the various important elements of “marketing mix” to deliver value to the customer.	10	2	1

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DEPARTMENT OF
INDUSTRIAL ENGINEERING AND MANAGEMENT

Date : 22 June 2024	**CIE – I**	**Max. Marks : 25**
Semester : IV	**UG**	**Duration : 60 Min**
Course Title: UNIVERSAL HUMAN VALUES		**Course Code : HS248AT**

Note:

1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
1.	Define self-exploration. What is the content of self-exploration?	5	3	1
2.	"Right understanding + Relationship = Mutual fulfilment; Right understanding + Physical facilities = Mutual prosperity." Illustrate the above with two examples for each.	8	3	3
3.	How can we ensure harmony in self ('I')?	4	1	4
4.	How do prosperity and wealth differ? What is more acceptable to us and why?	8	2	2

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DEPARTMENT OF
INDUSTRIAL ENGINEERING AND MANAGEMENT

Date : 07th July 2023	**CIE – I**	**Max. Marks : 25**
Semester : IV	**UG**	**Duration : 60 Min**
Course Title: UNIVERSAL HUMAN VALUES		**Course Code : 21HSU48**

Note:

1. Answer all the Questions.

Sl. No	Questions	M	BT	CO
1.	Distinguish between happiness and unhappiness?	5	4	3
2.	How to achieve harmony with the body?	5	1	1
3.	What is prosperity? Is it different from happiness?	5	2	1
4.	What is value education? Why there is a need of value education? How does value education help in fulfilling one's aspirations?	5	1	2
5.	What is the difference between prosperity and wealth? What is more acceptable to us and why?	5	3	4

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Department of Mechanical Engineering

Date: 02/04/2025	**CIE – Test 1**	**Max. Marks: 10 + 50**
Semester: IV	**UG**	**Duration: 120 Min**
Course Title: Material Science for Engineers		**Course Code: ME242TB**

S No	Questions	M	BT	CO
	Part – A			
1	According to which principle the electrons fill the lowest energy orbital first before moving to higher energy ones.	1	1	1
2	In _____ material the conduction and valence bands overlap so electrons can easily pass into the conduction band.	1	1	1
3	Define unit cell.	1	1	1
4	Write the classification of composites based on matrix.	2	1	1
5	The Burgers Vector is parallel to the dislocation line, the type of line defect is _____.	1	1	1
6	If an atom is missing from its normal site in the matrix, the defect is called _____ defect.	1	1	2
7	Define thermal conductivity with equation.	2	2	2
8	In dielectric materials the movement of electrons in response to an external electric field is _____ type of polarization mechanism.	1	1	2
	Part – B			
1(a)	With the help of neat sketches explain the Primary Interatomic bonds with examples.	10	2	1
2(a)	Classify solid materials explain them briefly with examples.	10	2	1
3(a)	Calculate Atomic packing factor for FCC unit cell.	5	3	2
3(b)	Classify and explain different crystal imperfections.	5	2	2
4(a)	Compare the properties and applications of metals and polymers.	10	3	1
5(a)	Illustrate the following thermoelectric effects: i) Seebeck effect ii) Peltier effect.	5	3	2
5(b)	Discuss different types of Insulating materials highlighting their applications.	5	2	2

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Department of Biotechnology

Date: 2nd April 2025	**Closed Book Offline Quiz & Test – 1**	**Max. Marks: 60**
Semester: IV	**UG**	**Duration: 120 Min**
Course Title: Biosafety Standards and Ethics (Basket Course)		**Course Code: BT242TC**

S No	Questions	M	BT	CO
	Quiz			
1	What is the role of the Institutional Biosafety Committee (IBC) in India?	1	2	2
2	The Recombinant DNA Advisory Committee (RCGM) is a part of which Department of Govt of India?	1	1	2
3	The committee of an Institutional Biosafety Committee (IBC) comprise of _____ and _____.	1	1	2
4	Biosafety levels indicate _____.	1	1	1
5	_____ is the primary mode of transmission of many viral biohazards.	1	1	1
6	Which PPE is essential for BSL 4 labs?	1	2	1
7	In class II biosafety cabinet, which direction does the air flow?	1	2	2
8	The Cartagena Protocol was adopted in which year?	1	1	2
9	Mention the 3 types of mechanism which takes place in HEPA filters.	2	3	2
	Test			
1	Comment on various types of Biohazards, and their impact on human health.	10	2	1
2	Explain Biosafety level II class B2 with a neat diagram.	10	2	1
3	Discuss the importance of biosafety guidelines in India and how they contribute to public health and safety.	10	3	2
4	Farmers in the Northern part of India are protesting against the genetically modified Mustard seeds. In this context discuss in detail the pros and cons of the GM Mustard seeds.	10	3	2
5	How do the Biosafety guidelines of the Government of India align with international biosafety standards? Discuss any significant differences or similarities.	10	2	2

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Department of Civil Engineering

Date: 02/04/2025	**CIE – I (Quiz + Test)**	**Max. Marks: 50**
Semester: III	**UG**	**Duration: 90 Min + 20 Min**
Course Title: Environment and Sustainability (Basket Course)		**Course Code: CV232AT**

S No	Questions	M	BT	CO
	Part – A			
1	List the components of environment.	2	1	1
2	Define biodiversity.	2	2	1
3	Differentiate between primary and secondary pollutants.	2	3	2
4	The permissible day and night noise levels for a silent zone in India are _____ dB and _____ dB respectively.	2	2	2
5	List any four sources of solid waste.	2	2	1
	Part – B			
1	Classify different types of eco system and explain any two briefly.	10	2	1
2	Define solid waste management and list the objectives of solid waste management.	10	2	2
3	Briefly explain the following safety terminologies. (i) Accident (ii) Ergonomics (iii) First aid (iv) Hazard (v) Safety inspection	10	2	2
4	Define noise pollution. List the effects of noise pollution and write the control measures.	10	2	2
5	With a neat sketch explain carbon cycle.	10	2	1

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Department of Biotechnology

Date: 05th July 2023	**CIE – 1**	**Max. Marks: 25**
Semester: IV	**UG**	**Duration: 60 Min**
Course Title: Environmental Technology		**Course Code: 21BT42A**

S No	Questions	M	BT	CO
1	State the SDG's promulgated by United Nations.	5	2	1
2	Discuss in brief the different components which constitute the environment.	5	3	1
3	State the significance of environmental education and NGO's in conserving the environment.	5	2	1
4	Explain the step by step process of Environmental Impact Assessment with flow chart.	5	2	1
5	Discuss the objectives and applications of ISO 14000 series in Environmental Management System (EMS).	5	2	1